

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

YEAR 11 PRELIMINARY EXAMINATION

SEPTEMBER 2007

Time Allowed: 120 minutes

Direction to Candidates:

- Approximately marks are shown alongside each question
- All necessary working should be shown. Marks may not be awarded for careless or badly arranged work
- Begin answering each question on a new page

Name: _____

Teacher: _____

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
/10	/11	/11	/11	/11	/11	/10	/11	/86

QUESTION 1 (10 marks)

a) Simplify $5\sqrt{2} - \sqrt{32}$ 1

b) Solve for x , 2

$$|x + 1| = 3$$

c) State the domain of $y = \sqrt{x - 1}$ 1

d) Solve for x : 2

$$\frac{2x}{3} - 1 = \frac{x+1}{4}$$

e) (i) Sketch the graph of $y = |x + 1|$ 1

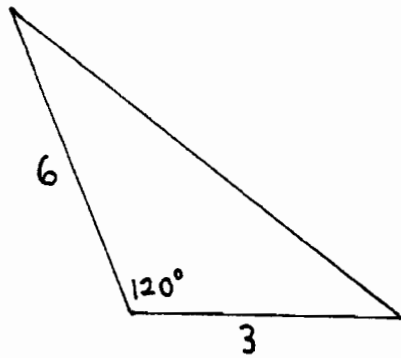
(ii) State its range 1

f) Find $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ 2

QUESTION 2 (11 marks)

a) If $\sin\theta = \frac{3}{7}$ and $0^\circ \leq \theta \leq 90^\circ$, find $\cos\theta$ in surd form 2

b) Find the area of the triangle below leaving your answer in surd form. 2



c) Sketch the region given by: $(x - 2)^2 + (y + 3)^2 > 9$ 3

d) Simplify $\frac{4a+2b}{8a+4b}$ 2

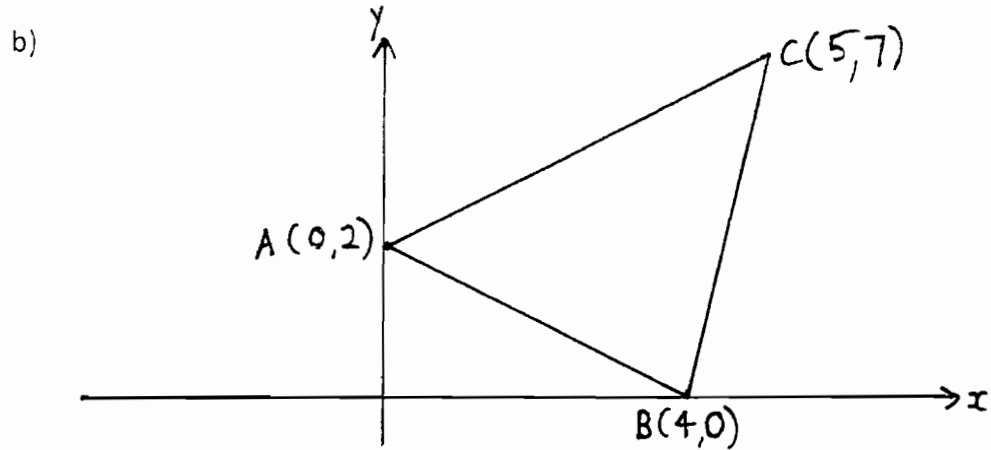
e) $f(x) = \begin{cases} x^3 + 1 & \text{if } x > 2 \\ 2x & \text{if } -1 \leq x \leq 2 \\ 5 & \text{if } x < -1 \end{cases}$ 2

Find $f(-2) - f(3) + f(2)$

QUESTION 3 (11 marks)

- a) Find the perpendicular distance from the point $(3,2)$ to the line $3x - 4y + 7 = 0$

2



- i) Find the gradient of AB 1
- ii) Find the coordinates of D , the midpoint of AB 1
- iii) Find the equation of the line passing through D and perpendicular to AB 2
- iv) Show that C lies on this line 1
- v) Find the lengths of AB and CD in surd form. 2
- vi) Find the area of the quadrilateral $ACBO$ 2

QUESTION 4 (11 marks)

- a) Solve $|2x - 1| < 3$ 2
- b) Write as a single fraction $\frac{1}{x-3} + \frac{1}{x+3}$ 2
- c) Solve $2 \sin \theta = -1$ for $0^\circ \leq \theta \leq 360^\circ$ 2
- d) Simplify $\cos \theta + \cos \theta \tan^2 \theta$ 3
- e) Prove that $\sec^2 \theta = \frac{1}{(1 - \sin \theta)(1 + \sin \theta)}$ 2

QUESTION 5 (11 marks)

a) Differentiate

i) $-3x^4$ 1

ii) $\frac{2x-1}{x+4}$ 2

iii) $(3x^2 - 5)^6$ 2

iv) $(2x + 3)(x^2 + x + 1)$ 2

b) Find the x co-ordinate of the point on the curve $y = x^2 + 2$
where the tangent has the gradient of -2 1

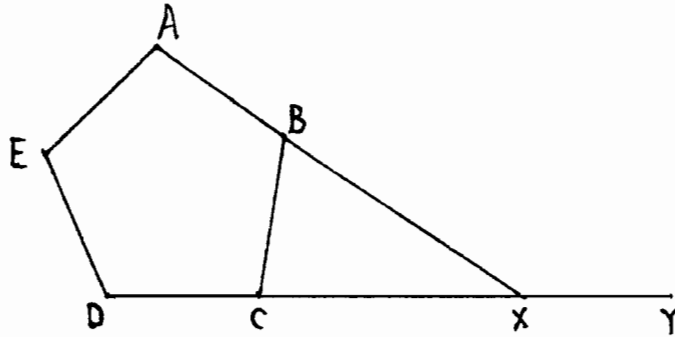
c) Find the equation of the tangent to $y = 2x^2 - 2x + 1$ at the point $x = 1$ 3

QUESTION 6 (11 marks)

a) Factorise $8 - 27x^3$

2

b)



In the diagram, $ABCDE$ is a regular pentagon and AB and DC are produced to meet at X . The point Y lies on DCX produced.

(i) Find $\angle ABC$

2

(ii) Find $\angle BXY$ giving reasons

3

c) (i) Find the discriminant of $x^2 + (k - 1)x + 1$ in simplest form

2

(ii) Find the range of values of k for which the quadratic expression above is positive definite.

2

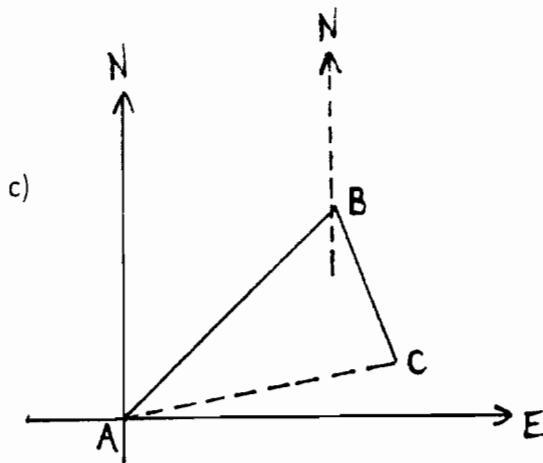
QUESTION 7 (10 marks)

- a) (i) Express the equation of the parabola $8y = x^2 - 8x - 24$ in the form $(x - h)^2 = 4a(y - k)$ 2
- (ii) Write down the coordinates of the vertex and equation of the directrix for this parabola 2
- b) If α and β are the roots of the quadratic equation $x^2 + 3x - 5 = 0$, find
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\alpha^2 + \beta^2$ 2
- (iv) $\alpha^3\beta + \alpha\beta^3$ 2

QUESTION 8 (11 marks)

a) If $f(x) = \frac{x-1}{x+2}$, show that $f(1-x) = \frac{x}{x-3}$ 2

b) Solve $9^x + 3 \cdot 3^x - 18 = 0$ by first reducing this equation to a quadratic 3



Copy this diagram onto your answer sheet and mark the following information on it.

- (i) An ultralight plane is flown from an airport A on a bearing of $030^\circ T$ for 150km to a position B. From position B the ultralight is then flown 200km on a new course bearing $135^\circ T$ to position C. Use the above diagram to find how far (to the nearest km) C is from A. 3

- (ii) Use the Sine Rule to help find the bearing of C from A. 3

(nearest degree)

Teacher's Name:

Student's Name/N°:

Solution to 2007 2 Unit Final Prelim. Exam

Question 1

$$a) \frac{5\sqrt{2} - \sqrt{32}}{5\sqrt{2} - 4\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2}} \quad (1)$$

$$b) |x+1| = 3$$

$$x+1=3 \quad \text{or} \quad x+1=-3$$

$$(1) \quad x=2 \quad \text{or} \quad x=-4 \quad (1)$$

$$c) x \geq 1 \quad (1)$$

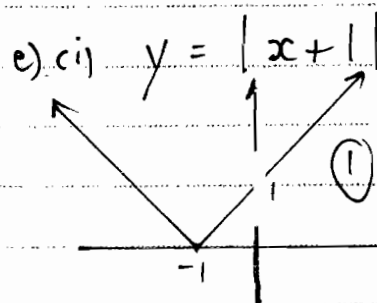
$$d) \frac{2x}{3} - 1 = \frac{x+1}{4}$$

$$8x - 12 = 3(x+1)$$

$$8x - 12 = 3x + 3 \quad (1)$$

$$5x = 15$$

$$x = 3 \quad (1)$$



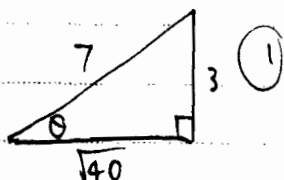
$$f) \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)} \quad (1)$$

$$= 4 \quad (1)$$

ii) Range $y \geq 0 \quad (1)$

Question 2

$$a) \sin \theta = \frac{3}{7}$$



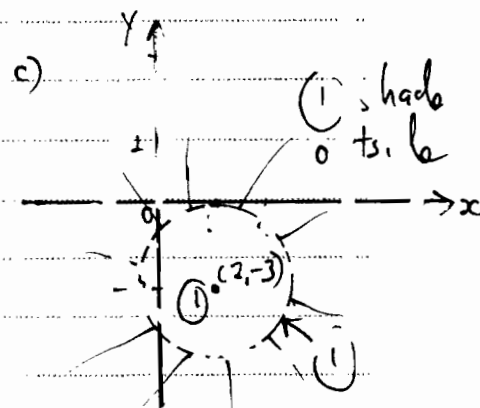
$$\cos \theta = \frac{\sqrt{40}}{7} \quad (1)$$

$$b) A = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 3 \times 6 \times \sin 120 \quad (1)$$

$$= 9 \times \frac{\sqrt{3}}{2}$$

$$= \frac{9\sqrt{3}}{2} \text{ units}^2 \quad (1)$$



$$d) \frac{4a+2b}{8a+4b}$$

$$\frac{1 \times (2a+b)}{2 \times (2a+b)} \quad (1)$$

$$= \frac{1}{2} \quad (1)$$

$$e) f(-2) - f(3) + f(2)$$

$$5 - 28 + 4 \quad (1)$$

$$= -19 \quad (1)$$

Question 3

$$a) d = \frac{|3 \times 3 + -4 \times 2 + 7|}{\sqrt{3^2 + (-4)^2}} \quad (1)$$

$$b) \text{ i) } m_{AB} = -\frac{1}{2} \quad (1)$$

$$\text{ ii) } D(2, 1) \quad (1)$$

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$$\text{ciii) } m = 2 \quad (2, 1)$$

$$y - 1 = 2(x - 2) \quad (1)$$

$$y - 1 = 2x - 4$$

$$y = 2x - 3 \text{ or } 2x - y - 3 = 0$$

$$\text{civ) } 7 = 2 \times 5 - 3$$

$$7 = 10 - 3 \quad (1)$$

$$\text{cv) } d_{AB} = \sqrt{(4-0)^2 + (0-2)^2}$$

$$= \sqrt{20} \text{ or } 2\sqrt{5} \quad (1)$$

$$\text{cvi) } \text{Area} = \frac{1}{2} \times 4 \times 2 + \frac{1}{2} \times 2\sqrt{5} \times 3\sqrt{5} \quad (1)$$

$$= 4 + 15$$

$$= 19 \text{ units}^2 \quad (1)$$

$$d_{CD} = \sqrt{(5-2)^2 + (7-1)^2}$$

$$= \sqrt{9 + 36}$$

$$= \sqrt{45} \text{ or}$$

$$= 3\sqrt{5} \quad (1)$$

Question 4

$$\text{a) } |2x - 1| < 3 \quad (1)$$

$$2x - 1 < 3 \quad \text{and} \quad 2x - 1 > -3$$

$$2x < 4 \quad \text{and} \quad 2x > -2$$

$$x < 2 \quad \text{and} \quad x > -1$$

$$-1 < x < 2 \quad (1)$$

$$\text{b) } \frac{1}{x-3} + \frac{1}{x+3}$$

$$\frac{x+3 + x-3}{(x-3)(x+3)} \quad (1)$$

$$= \frac{2x}{(x-3)(x+3)} \text{ or } \frac{2x}{x^2 - 9} \quad (1)$$

$$\text{c) } 2\sin\theta = -1$$

$$\sin\theta = -\frac{1}{2} \quad (1)$$

Working angle is 30°

$$\therefore \theta = 210^\circ, 330^\circ \quad (1)$$

$$\text{d) } \cos\theta + \cos\theta \tan^2\theta$$

$$\cos\theta(1 + \tan^2\theta) \quad (1)$$

$$\cos\theta \times \sec^2\theta \quad (1)$$

$$= \sec\theta \quad (1)$$

$$\text{e) } \sec^2\theta = \frac{1}{(1 - \sin\theta)(1 + \sin\theta)}$$

$$= \frac{1}{1 - \sin^2\theta} \quad (1)$$

$$= \frac{1}{\cos^2\theta} \quad (1)$$

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Question 5

a) (i) $-3x^4$

$$\frac{d}{dx} = -12x^3 \quad \textcircled{1}$$

(ii) $\frac{2x-1}{x+4}$

$$\frac{d}{dx} = \frac{(x+4) \cdot 2 - (2x-1)}{(x+4)^2}$$

$$\frac{d}{dx} = \frac{1}{(x+4)^2} \quad \textcircled{1}$$

(iii) $(3x^2-5)^6$

$$\frac{d}{dx} = 6(3x^2-5)^5 \cdot 6x$$

$$\frac{d}{dx} = 36x(3x^2-5)^5 \quad \textcircled{1}$$

(iv) $(2x+3)(x^2+x+1)$

$$\begin{aligned} \frac{d}{dx} &= (2x+3)(2x+1) + (x^2+x+1) \cdot 2 \\ &= 4x^2 + 2x + 6x + 3 + 2x^2 + 2x + 2 \end{aligned}$$

$$\frac{d}{dx} = 6x^2 + 10x + 5 \quad \textcircled{1}$$

b) $y = x^2 + 2$

$$\frac{dy}{dx} = 2x = -2$$

$$\therefore x = -1 \quad \textcircled{1}$$

c) $y = 2x^2 - 2x + 1$

$$\frac{dy}{dx} = 4x - 2$$

At $x=1$, $m=2 \quad \textcircled{1}$

At $x=1$, $y=1 \quad \textcircled{1}$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 1) \text{ or}$$

$$y = 2x - 1 \quad \textcircled{1}$$

Question 6

a) $8 - 27x^3 = 2^3 - (3x)^3 \quad \textcircled{1}$

$$(2-3x)(4+2 \cdot 3x+9x^2)$$

$$(2-3x)(4+6x+9x^2) \quad \textcircled{1}$$

b) Sum of Ext. Angles is 360

$$\therefore \text{Each exterior is } \frac{360}{5} = 72^\circ \quad \textcircled{1}$$

$$\therefore \angle ABC = 180 - 72$$

$$= 108^\circ \quad \textcircled{1}$$

(ii) $\angle BXY = \angle CBX + \angle BCX$

(Exterior angle) $\textcircled{1}$

$$\angle CBX = \angle BCX = 72^\circ \text{ from part (i) } \textcircled{1}$$

$$\therefore \angle BXY = 144^\circ \quad \textcircled{1}$$

c) (i) $x^2 + (k-1)x + 1$

$$\Delta = b^2 - 4ac$$

$$= (k-1)^2 - 4 \cdot 1 \cdot 1 \quad \textcircled{1}$$

(ii) Positive Definite

if $\Delta < 0$

$$k^2 - 2k - 3 < 0 \quad \textcircled{1}$$

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Question 7

$$\begin{aligned}
 \text{a) (i)} \quad x^2 - 8x - 24 &= 8y \\
 x^2 - 8x + 16 &= 8y + 16 + 24 \quad \textcircled{1} \\
 (x-4)^2 &= 8y + 40 \\
 (x-4)^2 &= 8(y+5) \quad \textcircled{1}
 \end{aligned}$$

(ii) Vertex is $(4, -5)$ $\textcircled{1}$
 Focal Length is 2
 Concave up $\therefore$ directrix
 is $y = -7$ $\textcircled{1}$

$$\text{b) } x^2 + 3x - 5 = 0$$

$$\begin{aligned}
 \text{(i)} \quad \alpha + \beta &= \frac{-b}{a} \\
 &= -3 \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \alpha \beta &= \frac{c}{a} \\
 &= -5 \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \quad \textcircled{1} \\
 &= (-3)^2 - 2 \times -5 \\
 &= 9 + 10 \\
 &= 19 \quad \textcircled{1}
 \end{aligned}$$

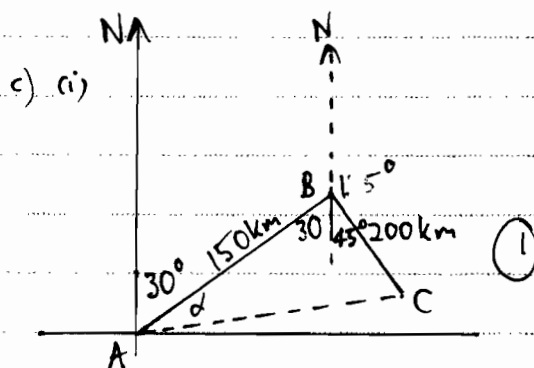
$$\begin{aligned}
 \text{(iv)} \quad \alpha^3 \beta + \alpha \beta^3 \\
 &= \alpha \beta (\alpha^2 + \beta^2) \quad \textcircled{1} \\
 &= -5 \times 19 \\
 &= -95 \quad \textcircled{1}
 \end{aligned}$$

Question 8

$$\text{a) } f(x) = \frac{x-1}{x+2}$$

$$\begin{aligned}
 f(1-x) &= \frac{1-x-1}{1-x+2} \quad \textcircled{1} \\
 &= \frac{-x}{3-x} = \frac{x}{x-3} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } 9^x + 3 \cdot 3^x - 18 &= 0 \\
 (3^2)^x + 3 \cdot 3^x - 18 &= 0 \\
 (3^x)^2 + 3 \cdot 3^x - 18 &= 0 \\
 \text{Let } v &= 3^x \\
 v^2 + 3v - 18 &= 0 \quad \textcircled{1} \\
 (v-3)(v+6) &= 0 \\
 v &= 3 \quad \text{or} \quad -6 \quad \textcircled{1} \\
 \therefore 3^x &= 3 \quad \text{or} \quad \cancel{3^x = -6} \\
 \therefore x &= 1 \quad \textcircled{1} \quad \text{No Sol'n}
 \end{aligned}$$



$$\begin{aligned}
 AC^2 &= 150^2 + 200^2 - 2 \times 150 \times 200 \cos 75^\circ \quad \textcircled{1} \\
 AC^2 &= \dots \dots 4 \\
 AC &= \dots \text{ km} \quad \textcircled{1}
 \end{aligned}$$

(ii) Need α

$$\begin{aligned}
 \textcircled{1} \quad \frac{\sin \alpha}{00} &= \frac{\sin 75}{1.7} \\
 \sin \alpha &= \frac{00 \times \sin 75}{1.7}
 \end{aligned}$$

$\alpha = \dots^\circ \quad \textcircled{1}$
 $\therefore$ Bearing of C from A
 is $30 + 38 = 068^\circ$ or